

Review and midterm prep

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Anthropology 215

210: HUMAN BIOLOGY

MIDTERM EXAM
16 October 2025
8.30-9.30AM

INSTRUCTIONS

Answer all questions on the answer sheet.
You may not turn the page and start the exam until instructed that you may do so by the invigilator.

STATIONERY REQUIREMENTS

- Black or blue pen
- Answer sheet

- 30 MCQ and T/F questions. Only 1 correct answer for each question.
- PLUS two optional short-answer bonus questions (worth 2 pt each)
- Closed-book exam

Final
will have
mandatory ones
so this is good practice

210: HUMAN BIOLOGY	MIDTERM EXAM	ANSWER SHEET
NAME		
EMAIL		

INSTRUCTIONS

Clearly indicate your answer by colouring in, crossing, or ticking in the box next to the answer you consider correct in black or blue ink, e.g.,

- ☐ b.
- ☒ b.
- ☒ b.

ANSWERS

- | | | | | |
|---|-----------------------------|-----------------------------|-----------------------------|-----------------------------|
| 1 | ☐ a. | ☐ b. | ☐ c. | ☐ d. |
| 2 | ☐ a. | ☐ b. | ☐ c. | ☐ d. |
| 3 | ☐ a. | ☐ b. | ☐ c. | ☐ d. |
| 4 | ☐ a. | ☐ b. | ☐ c. | ☐ d. |

On the day itself

Before the exam starts

- The room will be prepped with exam papers and answer sheets
- Please come on time: the exam will start promptly.
- Leave bags etc. at the front of the room and take a seat

During the exam

- If you finish early, please bring your exam paper and answer sheet to the invigilator, take your bag quietly, and leave quietly

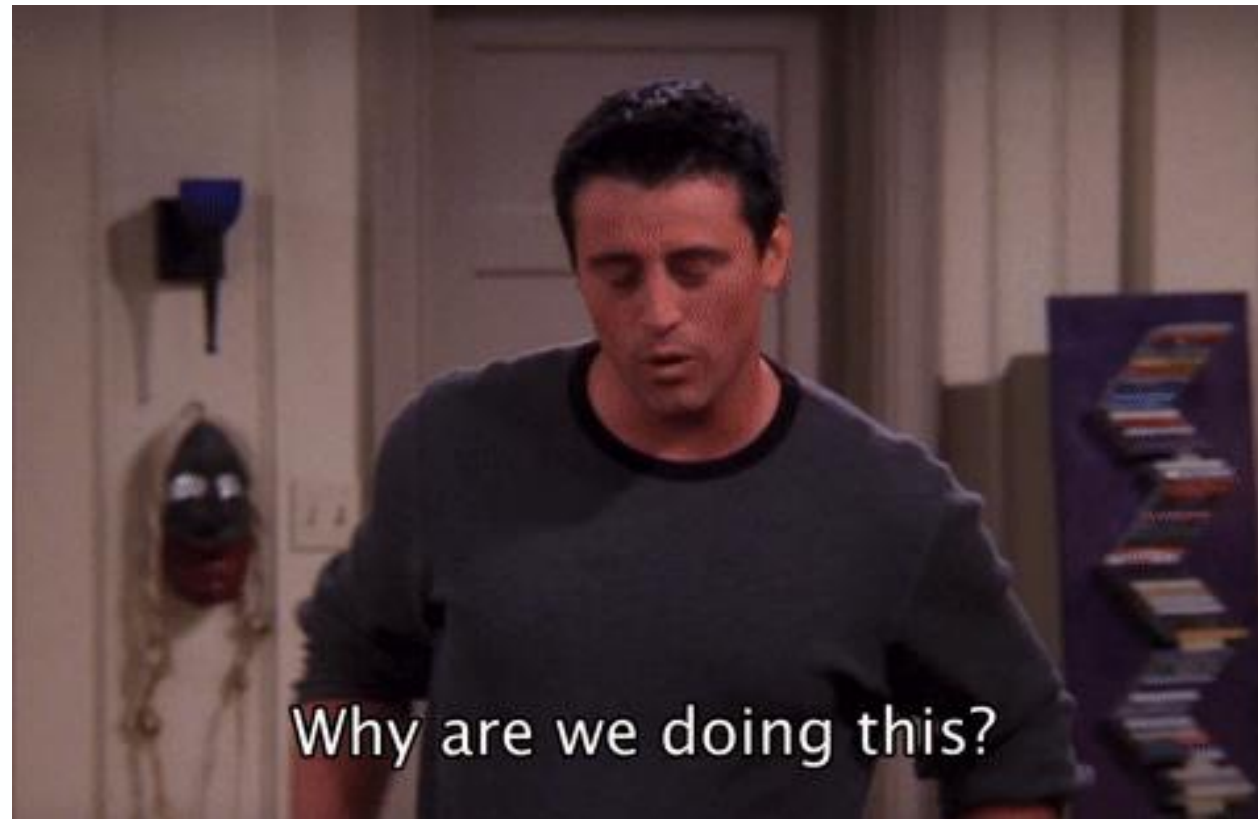
When the exam finishes

- When the invigilator ends the exam, stay seated until your exam paper and answer sheet have been collected

After the exam

- Grades will be posted on Canvas
- You will receive your marked answer sheet

Why a midterm?



Why a midterm?

To test

1. How much you're recalling from lectures (**not readings!**)
2. How much you're understanding this material

→ 540 score 3=3

What's on it?

- Nothing that is not in lectures
- So: know lectures 1-12
 - and deepen your knowledge with the readings *if you want*



Recalling: facts

- Can you define key concepts defined in the lectures?
- Can you recall key patterns, trends, and results?

1. Which of the following processes does not necessarily result in adaptation?

- a. Natural selection
- ☒ b. Ontogeny *just growing up*
- c. Plasticity
- d. Developmental plasticity

4. Fill in the blanks:

Need to know about this paper
Lewontin (1972) found that the average _____ across classical markers was _____. This means that _____ % of modern human variation is found across all human populations. *Need to understand this Working for a fst*

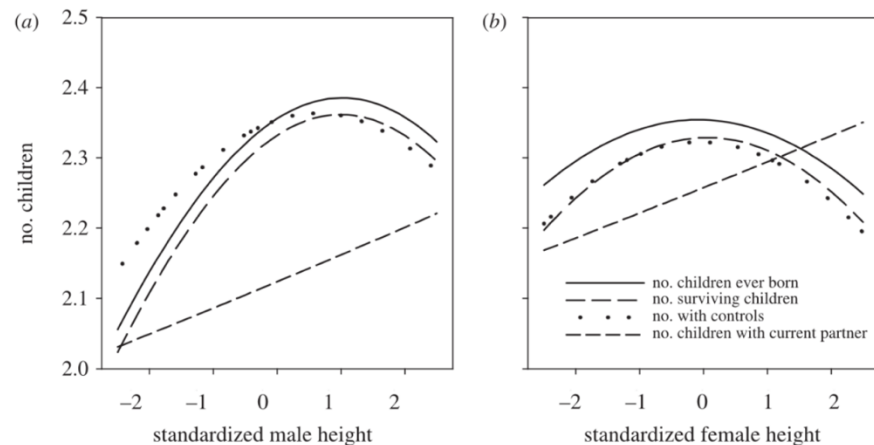
- a. Diversity, 0.85, 85
- b. F_{ST} , 0.15, 15
- ☒ c. F_{ST} , 0.15, 85
- d. Diversity, 0.15, 15%

How do you revise facts?

Understanding: applying and making connections

- Can you apply concepts to make sense of new information?
- Do you recognize concepts across fields/lectures?

2. In a study, researchers measured participants' height and the number of children they have, in order to ask whether natural selection could have been acting on this population. They found the following pattern:



What might the number of children be used for as a proxy in this study?

- ☒ a. Fitness
- b. Adaptation
- c. Population-level patterns of growth during childhood
- d. Natural selection

Look out for orange 'key' boxes in lectures!

Don't only study these but these are important

What are the common themes across lectures?

Frameworks

→ tradeoffs - tradeoffs can happen at both proximate + ultimate causes

- proximate vs ultimate

↳ Tinbergen's 4 questions to think about why something is the way it is

Stressors

↳ heat stress and disease

↳ Signal for disease

related positive selection

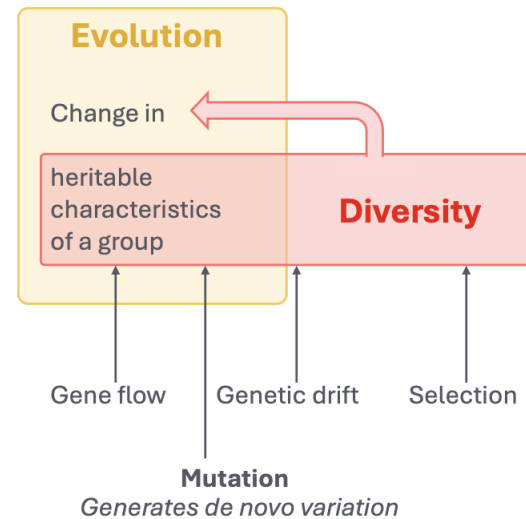
↳ culture + behavior can't buffer

against disease

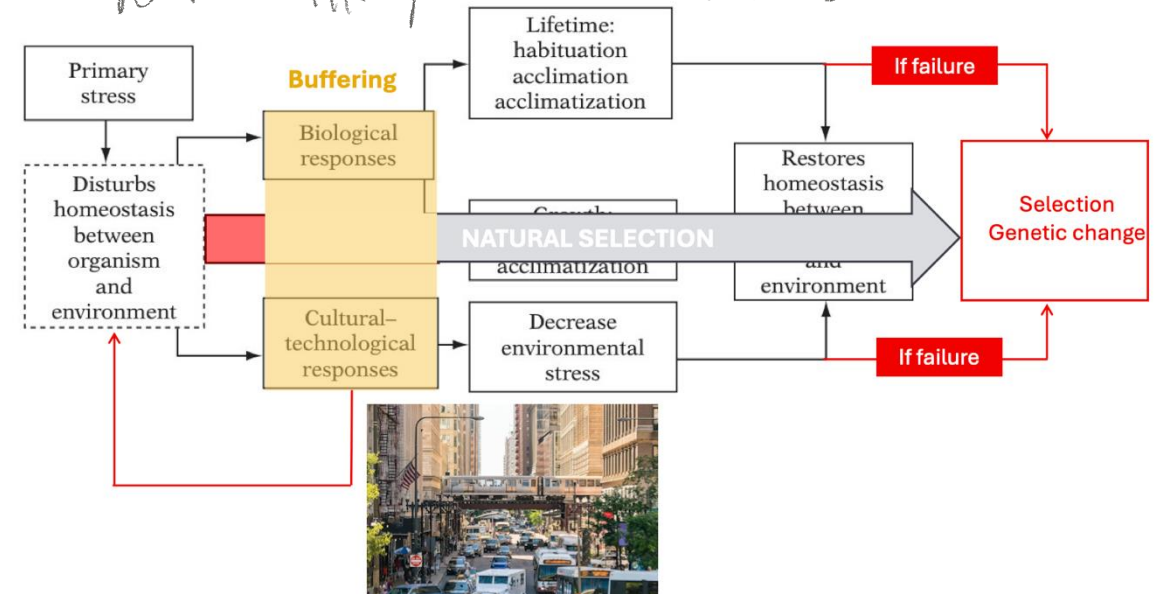
Group-level thinking

The big takeaways: Lectures 1-3

1. Use the big frameworks as handrails/anchors throughout the course
2. **Evolution is change, over time, in heritable characteristics of a group**
↳ Does not happen to level of individual. Individuals can adapt though
3. Genetic variation refers to differences between genomes; **diversity** is a term to describe (the amount of) **variation within and between groups**
4. **Evolution (and changes in diversity) happens through mutation and gene flow, drift, and selection**; these latter three change the frequencies of mutations.



Buffering is not a force or process. Group that describes non-genetic adaptive processes that prevent genetic change from happening (don't need natural selection)



The big takeaways: Lectures 4-5

Skin colour,
fur, scalp
hair*

Plasticity

Homo sapiens
autapomorphies

Locomotion

Foramen Magnum
- Inferior anterior to ramus
Mandible - associated w/
bipedalism?

7 Ma

6 Ma

5 Ma

4 Ma

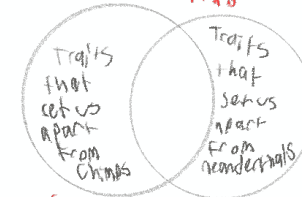
3 Ma

2 Ma

1 Ma

- Definition of hominins?
- Traits inherited from mammal, vertebrate, etc ancestors? - mammalian jaw teeth

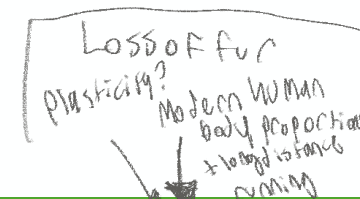
Looking at exclusively human
traits - Differentiate us from Neandertals
Must have evolved after split from
Neandertals



↳ Traits that set us apart
from Neandertals also
set us apart from chimps
(but not necessarily vice versa!)

plasticity - we can't really look
at plasticity in fossil record.
we do not know when human levels
of plasticity appeared

Best guess is 2 Ma based on
traits that covary w/ human
plasticity - Best guess is erectus,
which is similar to us



Invariant traits
What are invariant traits that are invariant
vs human? what are traits that are invariant
across humans + set us apart from
closest living relative? so had
to have evolved after split from Mon

Some examples compared to apes
Obligate terrestrial bipeds
↳ canines would have been reduced at times
↳ High encephalization quotient

↳ Hairlessness
↳ Loss of sweating
↳ Foramen magnum directly underneath

2 Ma is not when
terrestrial bipedalism
appears but when human
form of it tailored
to long distance running
appeared

Reptilian jaw - mammalian
ears and jaw - accurate
hearing. Inherited from
last common ancestor of
mammals

Hominids - ares more related
to us than to chimps and
such

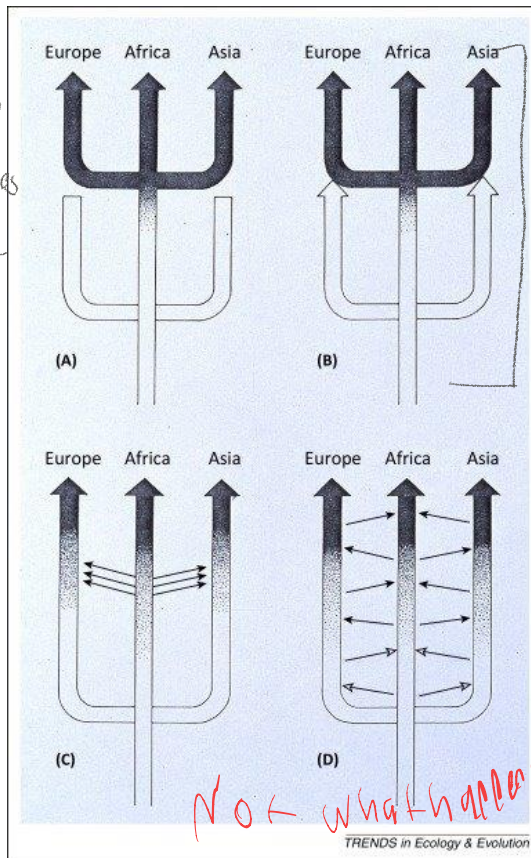
Important to conceptualize
time line of when things
evolved

The big takeaways: Lectu

Origins of
Homo sapiens

Evidence shows our species evolved
in Africa

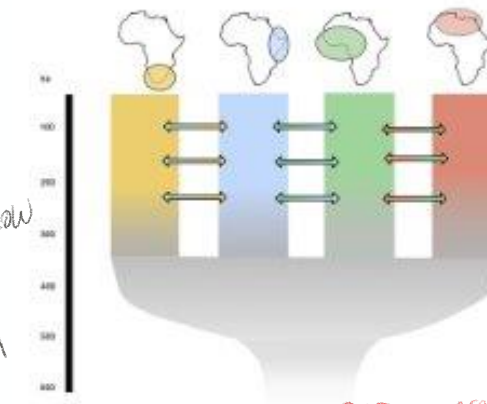
Multiple pops sometimes
in contact, sometimes
isolated. One pop that
expands out



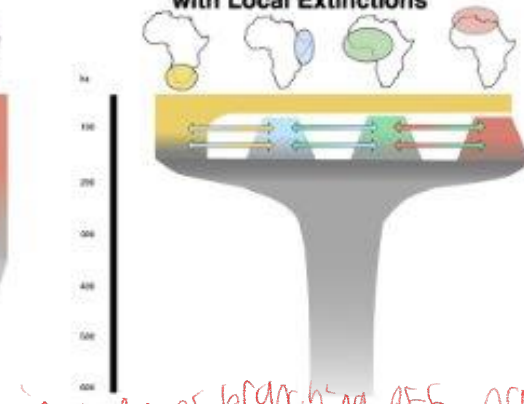
out of
Africa
w/ gene flow
w/ archaic
outside
of Africa

Not what happened!

(a) Model 1: African Multiregionalism



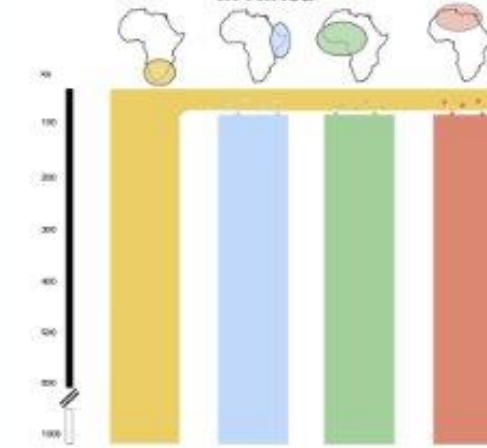
(b) Model 2: Single Origin Range Expansion with Local Extinctions



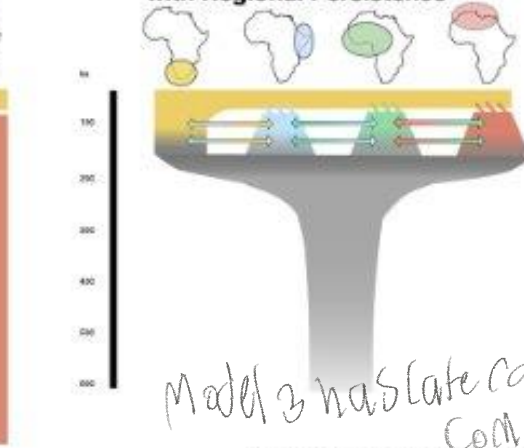
But probably not
a single pop in
Africa becoming human-
different pops that
admix when connected
and are isolated at other
times

Difference in time of branching off, nature
of gene

(d) Model 4: Archaic Hominin Admixture in Africa



(c) Model 3: Single Origin Range Expansion with Regional Persistence



of low, and
what happens
to other
pops

Model 3 has late gene flow
compared to model 2

7 Ma

6 Ma

5 Ma

4 Ma

3 Ma

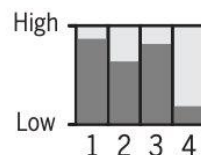
2 Ma

1 Ma

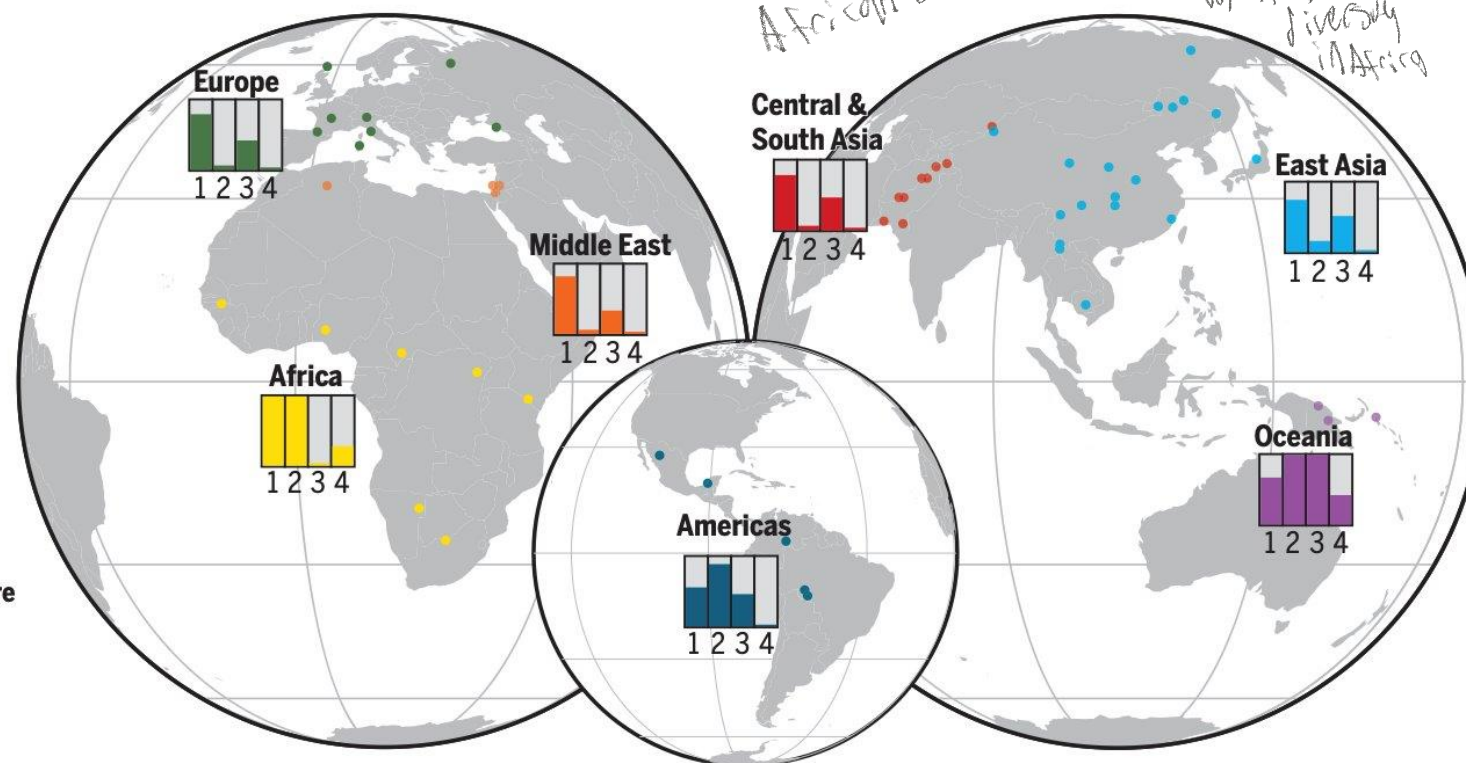


The big takeaways: Lectures 4-5

Amounts of different classes
of genetic variation



- 1. Total variation**
Relative to highest value (Africa)
- 2. Private, common variation**
Relative to highest value (Africa/Oceania)
- 3. Variation deriving from archaic admixture**
Relative to highest value (Oceania)
- 4. Private, common variation shared with archaic genomes**
Fraction of private variation



7 Ma

6 Ma

5 Ma

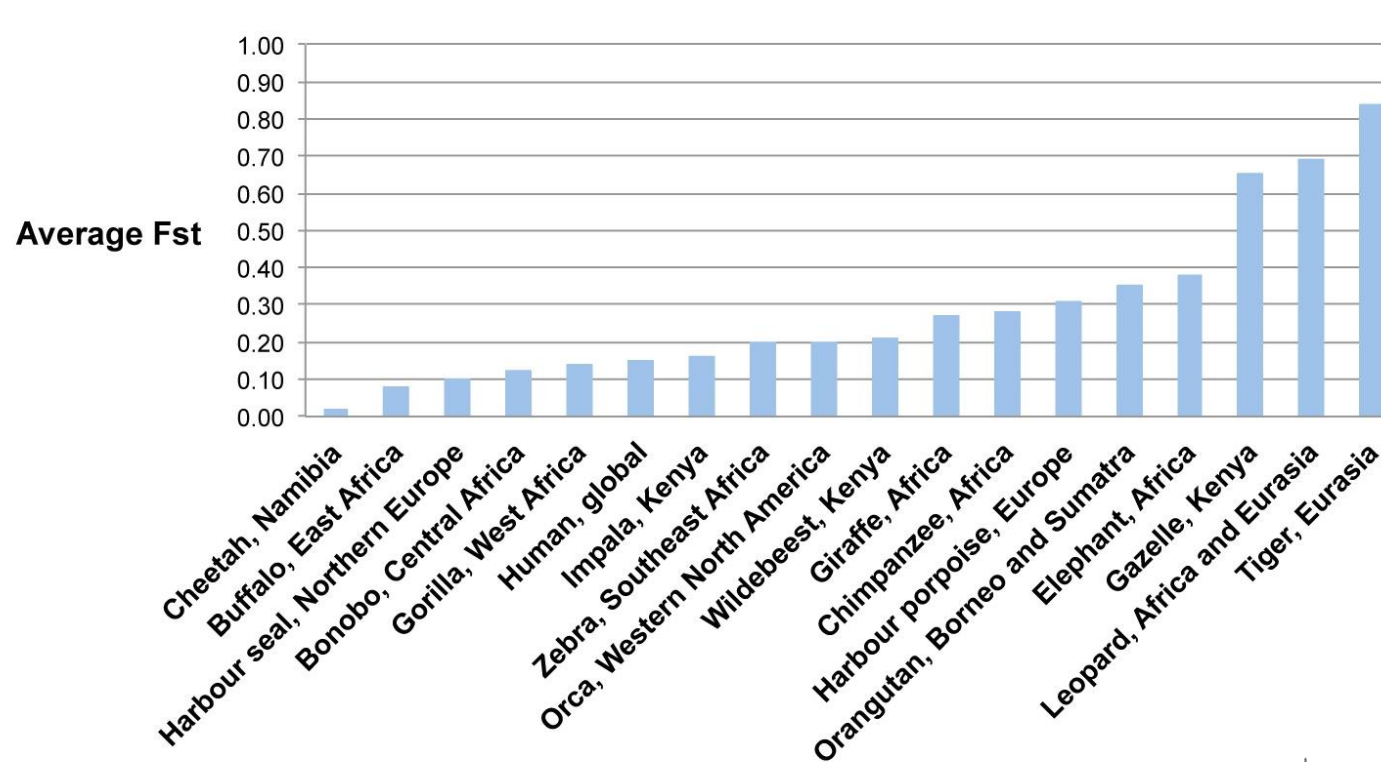
4 Ma

3 Ma

2 Ma

1 Ma

The big takeaways: Lectures 4-5



*
Recent origin w/
low effective
pop size
based on F_{ST}

small founder pop moving out from Africa

7 Ma

6 Ma

5 Ma

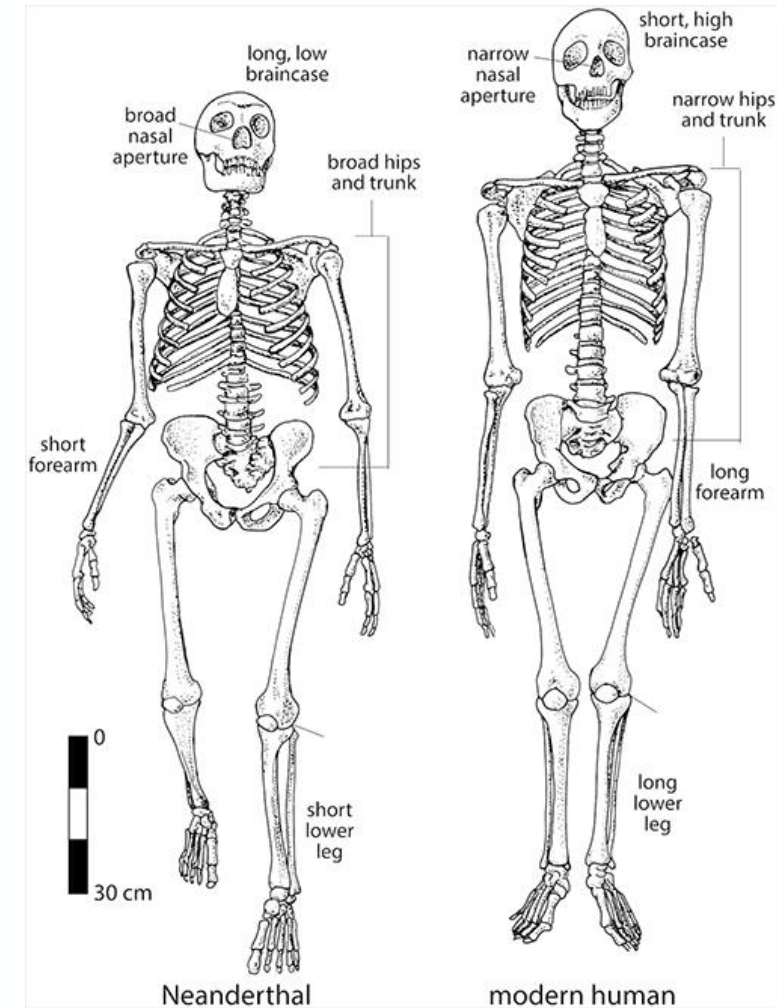
4 Ma

3 Ma

2 Ma

1 Ma

The big takeaways: Lectures 4-5



7 Ma

6 Ma

5 Ma

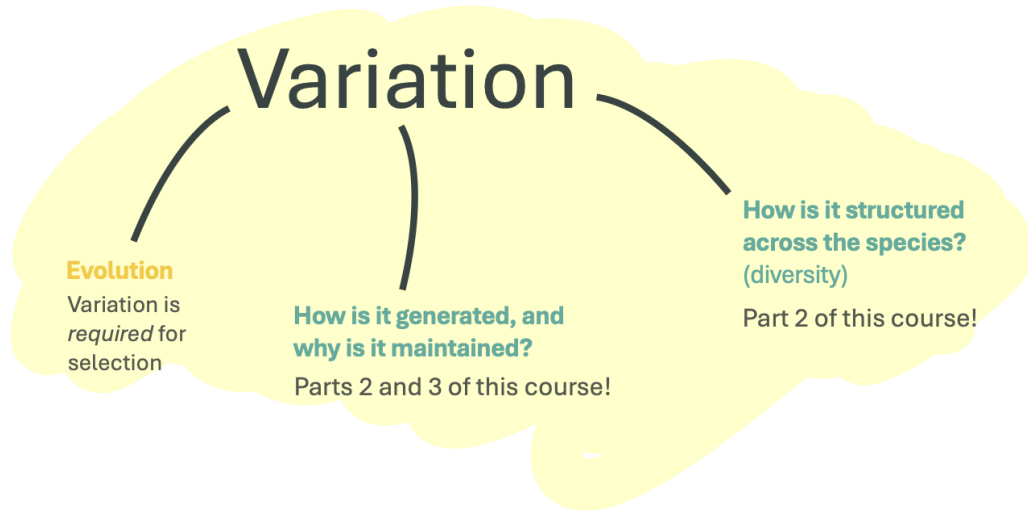
4 Ma

3 Ma

2 Ma

1 Ma

The big takeaways: Lecture 6



Modern best practices in ethics

- Modern ethical standards in many cases explicitly formulated in response to historical transgressions
 - E.g., informed consent: Nuremberg Code.
- Key: **Informed consent**, and the **weighing up of risks and benefits**
- **Involvement of study subjects in research design** (where appropriate)
- Culturally appropriate presentation of results to the scientific community *and* study subjects
- Capacity-building: including, promoting, supporting, and acknowledging full authorship of local researchers
 - Methods workshops on data collection and analysis
 - Sponsorship of university-level education (especially abroad)
 - Sponsorship of conference attendance

The big takeaways: Lectures 7-9

1. The human genome is (uniquely?) **buffered** from intense selection. Most human genetic variation is selectively neutral. Culture plays an important role, but there are also arguments that humans are exceptionally developmentally adaptable.
2. However... there are stressors that surpass short-term processes! One of the most important stressors (measured in the number of known genetic adaptations *and* the strength of selection) is **disease**.

- Lactase persistence
- Malaria resistance
- Short stature in RFHG

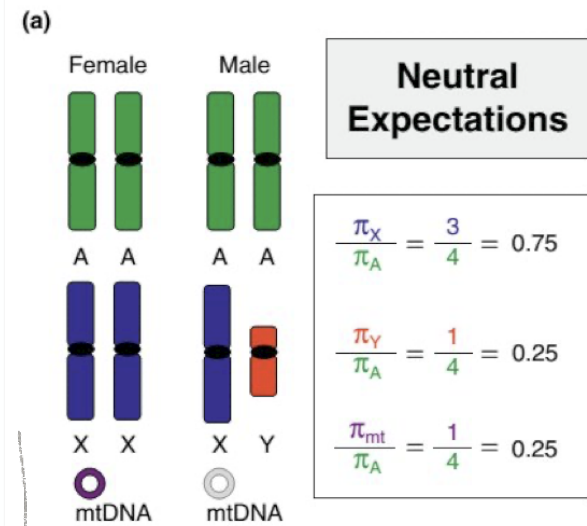
LP reflective
of disease rather
than diet
LNP - can't efficiently
digest lactose → stomach
upset
LNP people still drink milk,
not a matter of fitness
↳ Neither negative fitness of LNP
and no increased fitness for LNP
Currently we are very buffered culturally to
we can get whatever food we want
But in the past - crisis or chronic context
↳ see notes about this context

Nearly neutral - new variants don't need
to be super good or super bad

The big takeaways: Lectures 7-9

What is going on?

Key



Under random mating and neutrality, diversity of autosomes/ X/Y/mtDNA follow the **ploidy of transmissible copies** in the population

Neutral expectations:

- N_e or diversity follows a 4:3:1:1 ratio across the autosomes, X, Y, and mtDNA respectively
 - That is, autosomal diversity is expected to be 4x higher than mtDNA diversity; and Y and mtDNA are expected to be equally diverse

Sex-biased processes drive divergence from these expectations.

- Thus, comparing diversity of loci can provide a view into male vs female demography.

↑
No cultural influence
of effective male or female
population size

↑
Cultural
drives such as
marriage
practices and variance
in reproductive success
shift # of effective
pop size for a given sex

Webster and Wilson Sayres 2016 Curr Opin Genet Dev



Matrilocal: expect greater Y-chromosome diversity

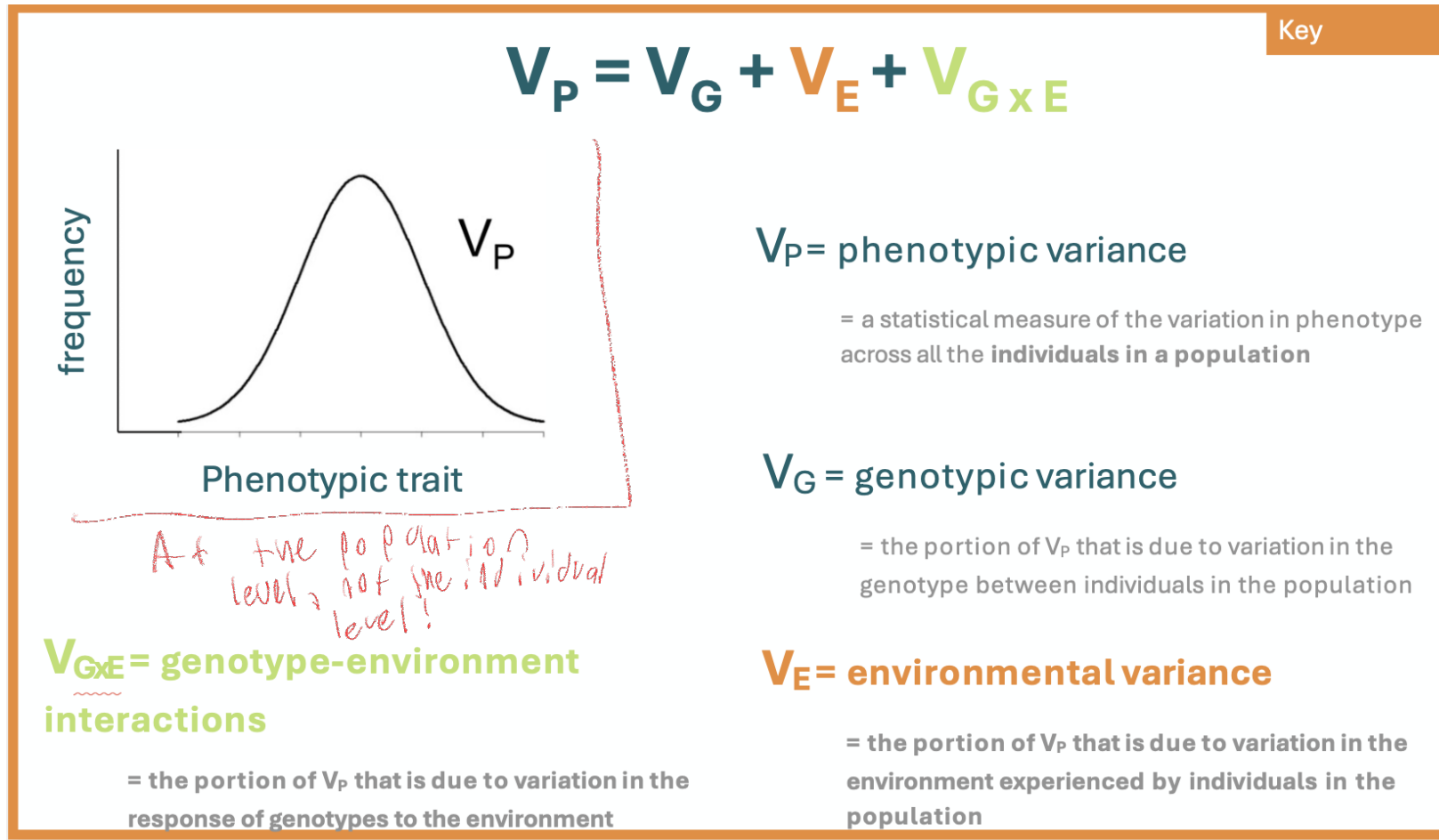
- mtDNA less diverse
- Y more diverse



Patrilocal: expect greater mtDNA diversity

- mtDNA more diverse
- Y less diverse

The big takeaways: Lecture 10



The big takeaways: Lecture 10

- Epigenetics
- Genomic imprinting
- Phenotypic plasticity/reaction norms

The big takeaways: Lectures 11-12

clinal variation

General patterns

- Males tend to be taller, have longer limbs vs trunk, and be heavier than females

- People are generally taller at higher latitudes
- People generally have more fat at higher latitudes
- People generally have shorter limbs at higher latitudes

- Pygmy phenotypes

General processes

Big role for genes, but lots of overlap between males and females, and a huge effect of environmental variables

Allen's Rule holds for limb lengths; recent data does *not* support Bergmann's rule.

In all cases, secular trends *and* specific studies show that nutrition during growth plays a major role.

Nutrition generally affects extremities most!

'Caused' by an early cessation of growth – not a decreased *rate* of growth. As for the ultimate reason... the jury is still out!

The big takeaways: Lectures 11-12

- Why did humans lose their fur?
- Why do humans have scalp hair that is so exceptionally diverse in colour *and* structure?
- How is scalp hair colour and structure distributed across the world? Why?

Categories
to capture
this diversity
is reflective
of actual
distributions

The *nearly* naked ape
with exceptional skin colour
diversity

- Why did humans lose their fur?
- Why do humans have skin that is so exceptionally diverse in colour?
- How is skin colour distributed across the world? Why?

Why does y chr diversity tend to be lower than x diversity at smaller scales?

Ploidy wise - 1 male + 2 females = 3 x chrs, 2 y chr

↳ Diversity follows ploidy

Y chr has more purifying selection

↳ purifying selection on both male chrs b/c both x and y are expressed → subject to selection

Variance in reproductive success

Timeline

Lots of traits evolved 2mya

AMH mosaic evolution w/ cranial traits → see notes abt